Carrol Astrophysics Chapter 1: The Celestial Sphere

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1 Geocentric Models of the Universe

Example: (Plato's Model of the Universe)

Plato theorirzed that celestial bodies were fixed on a sphere that rotated with a constant speed about the Earth. This means that all the stars should move with a constant speed across our night sky, but this was not the case.

Definition: (Celestial Poles and Equator)

The **celestial poles** are extensions of the north and south poles upward, "piercing" the heavens above. The **celestial equator** was also an extension of the equator upwards.

Definition: (Retrograde Motion)

When an object in the sky appears to suddenly change direction, it is called **retrograde motion**.

Example:

Mars exhibits retrograde motion at certain times.

Example: (Hipparchus's Model of the Universe)

To adress retrograde motion, Hipparchus proposed that most stars indeed were moving in a constant rotation, called the **deferent**, but that some stars had their own rotation about a point in the deferent, which was called their **epicycle**.

This system was improved by Ptolemy, who added the concept of an **equant**, which was the point from where the motion of a star in an epicycle appeared constant. He also moved the Earth away from the center of the deferent.

Example: (Copernicus's Discoveries)

Copernicus introduced the concept of a heliocentric model. By noticing that Mercury and Venus are never seen more than $28°$ and $47°$ degrees East or West of the Sun, we can deduce that they are orbiting closer to the Sun than the Earth is.

Definition: (Inferior Planets)

Inferior planets are planets that are orbiting closer to the Sun than the Earth. **Superior planets** are planets that are orbiting farther from the Sun than the Earth.

The **greatest Eastern/Western elongation** is the largest angle that a planet/object can be seen away from the Sun. For inferior planets, this will be some angle less than 90 degrees.

Superior planets can be seen in an alignment known as **opposition**, where they are on the opposite side of the sky as the Sun.

Inferior planets can be seen in **inferior conjunction**, when they are in between the Earth and the Sun

Definition: (Orbital Periods)

The **synodic period** of a celestial object is the time it takes for that object to return to the same configuration about the Sun.

The **sidereal period** of an object is the actual time it takes for an object to make a complete orbit, relative to the background stars (which are very far away and assumed to be pretty stationary).

Theorem:

If S is the synodic period of an object, P is the sidereal period of a celestial object, and $P_\oplus$ is the sidereal period of the Earth, the three are related by

$$\frac{1}{S} = \begin{cases} \frac{1}{P} - \frac{1}{P_\oplus} & \text{for inferior objects} \\ \frac{1}{P_\oplus} - \frac{1}{P} & \text{for superior objects} \end{cases}$$

2 Positions on the Celestial Sphere

Definition: (Altitude-Azimuth Coordinates)

The **altitude-azimuth coordinate system** assigns each celestial object two coordinates. First, consider a plane passing through the observer, the center of the Earth, and the star you are looking at. Along this plane, the **altitude** is the angle from the horizon to the star you are looking at, and the **azimuth** is the angle from the North pole to the plane (where East is positive).

The plane intersecting the North pole, the South pole, and the observer is called the **meridian**.

THe **zenith** is the angle between straight vertical for the observer, and the star they are looking at, in the altitude plane. Zenith and altitude should always add to $90°$

Definition: (Ecliptic)

By photographing the position of the Sun at midday every day, we can see that it see that it seems to move throughout the sky. This path is called the **ecliptic**.

Definition: (Time)

Solar time is the time it takes for the Sun to cross the meridian. Since the Earth is orbiting around the sun, on average, the Earth must rotate $361°$ in one solar day

Sidrereal time is the time it takes for distant celestial objects to cross the meridian.

Definition: (Equinoxes and Solstices)

Twice a year, the ecliptic crosses the celestial equator. When crossing from South to North, the location is called the **vernal equinox**. When crossing from North to South, it is called the **autumnal equinox**.

The northernmost point of the ecliptic is called the **summer solstice**. The southernmost point of the ecliptic is called the **winter solstice**.

Definition: (Equatorial Coordinates)

Equatorial coordinates are given by two values:

Declination, or δ, is measured in degrees North or South of the celestial equator.

Right ascension, or α, is measured East, from the vernal equinox to the object in question. It is given in hours, minutes, and seconds.

Remark:

Equatorial coordinates may experience some variation due to precession of the Earth's axis, and other wobbling, but overall it is pretty accurate.

Definition: (Sidereal Time)

Since the vernal equinox is stationary in the sky, and the Earth rotates under it, we can define **local sidereal time** to be the amount of time since the vernal equinox last crossed the meridian.

Definition: (Precession)

Precession is the wobbling of the Earth's axis of rotation due to the Earth not being a perfect sphere, and gravity from the moon and stars.

Example:

The precession of the Earth has a period of 25770 years.

Definition: (Epochs)

Precession will change the location of the vernal equinox, so we must talk about equatorial coordinates in the context of a specific **epoch**, or a commonly agreed upon time for when the position of the equinox is well known.

Proposition: (Accounting for Precession)

Changes in the location of the vernal equinox are given by the following equation:

$$\Delta\alpha = M + N \sin\alpha \tan\delta$$

$$\Delta\delta = N \cos\alpha$$

where

$$M = 1.2812323^\circ T + 0.0003879^\circ T^2 + 0.0000101^\circ T^3$$
$$N = 0.5567530^\circ T - 0.0001185^\circ T^2 - 0.0000116^\circ T^3$$

and

$$T = \frac{t - 2000}{100}$$

where t is the current Julian year.

Example: (Measurements of Time)

The **Julian date** is the amount of days that have elapsed since noon on January 1st, 4713 BC.

Heliocentric Julian date is the Julian time of events if they were seen from the center of the Sun.

Theorem: